

A Finite Source–Scale Ladder for Signed Prime–Shell Spectral Profiles

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September 1, 2026

Abstract

TPC–324 replicated a finite signed spectral-profile pattern at new source locations. We next ask whether the pattern survives a controlled change of source cardinality. Using the same literal deleted-diagonal centered prime-shell blocks, we freeze a new source origin and a nested four-rung ladder with source counts 160, 320, 640, 1280. Across $Q \in \{24, 36, 54, 80\}$ and exponents $s \in \{1, 2\}$, the all-plus coherent normalized profile majorizes the direct profile on all 32 rows. The outward lower envelope of profile total variation and the outward upper envelope of the coherent energy ratio both decrease strictly across the four rungs. These are finite numerical certificates and observations, not an asymptotic theorem, an arithmetic cancellation estimate, or a twin-prime result.

1 Question and claim boundary

Let $I_N = [12001, 12000 + N/2] \cap \mathbb{Z}$ for $N \in \{320, 640, 1280, 2560\}$. Thus the four actual source counts are $N/2 = 160, 320, 640, 1280$, and the intervals are nested. For a prime $p \in (Q, 2Q]$, define the literal block

$$B_{p,N}^{(s)}(u, t) = p \frac{H^{2s}}{(H^2 + (u - t)^2)^s} \left(\mathbf{1}_{p|u-t} - \frac{1}{p-1} \right) \mathbf{1}_{u \neq t} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid t}, \quad H = 66. \quad (1)$$

The direct and signed Grams are

$$G_0(N) = \sum_p B_p^* B_p, \quad G_e(N) = \left(\sum_p e_p B_p \right)^* \left(\sum_p e_p B_p \right). \quad (2)$$

We use four predeclared laws: all-plus, index-alternating, the mod-4 character, and half-split signs.

For a positive-trace Gram G , let

$$\pi(G) = \left(\frac{\lambda_1(G)}{\text{tr } G}, \dots, \frac{\lambda_n(G)}{\text{tr } G} \right), \quad \lambda_1 \geq \dots \geq \lambda_n.$$

We say that the signed profile majorizes the direct profile when every interior partial-sum difference is nonnegative and at least one is positive. The principal claim is deliberately finite:

NUMERICALLY CERTIFIED FINITE: all-plus majorization on 32/32 ladder rows, with a strict descending four-rung diagnostic envelope.

2 Exact structure and protocol

Every Gram above is of the form A^*A (or a finite sum of such matrices), so positive semidefiniteness and positive spectral-profile typing are exact finite facts (standard spectral facts are summarized in [1]). Also, for $a > 0$, $\pi(aG) = \pi(G)$ while $\text{tr}(aG) = a \text{tr}(G)$. Energy ratio and normalized shape are therefore different observables; one cannot be substituted for the other.

The scale intervention is source cardinality only: the origin, height, shell anchors, exponents, kernel, and sign menu are held fixed. The producer is provenance-locked to the TPC–324 literal engine. It records forward SciPy, forward NumPy, and reverse NumPy accumulation/spectral paths. A separate checker rebuilds the blocks in reverse order with an independent einsum implementation. A stress suite verifies strict nesting, disjointness from all prior frozen panels, a nontrivial residue-mask perturbation, the envelope trend, and the claim firewall.

3 Results

Table 1 gives the all-plus scale envelopes. Each rung contains eight rows (four shell anchors and two exponents). The prefix lower endpoint is positive at every rung; the two displayed envelope columns decrease strictly. The numbers are outward-rounded finite diagnostics.

Table 1: All-plus finite scale ladder.

N	$ I_N $	min prefix lower	min TV lower	max energy ratio
320	160	$1.25523 \cdot 10^{-5}$	0.285851	8.900245
640	320	$3.30131 \cdot 10^{-4}$	0.211205	6.868573
1280	640	$6.91139 \cdot 10^{-5}$	0.190206	6.250493
2560	1280	$1.64929 \cdot 10^{-5}$	0.170135	5.999045

The full profile census is

sign law	signed majorizes	mixed	direct majorizes
all-plus	32	0	0
alternating index	21	11	0
mod-4 character	26	6	0
half-split	23	9	0

The all-plus statement is thus stable over the declared ladder, while the alternative laws remain mixed rather than converging to a selected canonical arithmetic sign. The all-plus energy ratio is below one on 4/32 rows and above one on 28/32 rows. This crossing is compatible with, and reinforces, the amplitude/shape separation.

As a small exact anchor at the new origin, the interval $[12001, 12016]$ with $Q = 4, s = 1$ has rational direct and alternating-signed energy digests `b21611...982990` and `6dd9a7...4a5d6`, respectively. Their decimal energies are approximately 1476.329537716392 and 1631.802032195217. The complete numerators and denominators are retained in the canonical certificate rather than rounded into the manuscript.

4 Interpretation and obstruction

The result advances the route from source-location replication to a fixed-origin finite scale audit. It does not establish that the four-rung envelopes have a limit, or that a profile law is uniform for arbitrary source intervals. In particular, the nested intervals change the Hilbert-space dimension, so their comparison is not an identity under unitary translation. The imported literal engine also contains no Möbius or von Mangoldt weighting.

Accordingly the arithmetic status is

$$\text{ARITHMETIC_ADVANCE} = \text{NO}, \quad \text{FIXED_POWER_CREDIT} = 0, \quad \text{FULL_GATE_B} = \text{OPEN}.$$

The next minimal tests are a disjoint second scale ladder and a source-native signed arithmetic L^2 estimate. The Session-named official Route-A and Route-B evaluator files are absent from this checkout; the local Bridge-B record is therefore fail-closed and makes no official-pass claim.

5 Reproducibility

The repository package contains the frozen protocol, derivation and proof packages, canonical JSON certificate, independent replay, stress suite, and the local Bridge-B checker. From the repository root, run:

```
python -B papers/tpc-325-scale-ladder-profile/code/tpc325_scale_ladder_profile.py --check
python -B papers/tpc-325-scale-ladder-profile/experiments/tpc325_independent_checker.py --check
python -B papers/tpc-325-scale-ladder-profile/experiments/tpc325_scale_stress.py --check
```

The optimized Python counterparts and the Bridge-B normal/optimized equality check are part of the release contract.

References

- [1] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition, 2013.